

claude-windows / ac48091f-a4f4-4268-b968-75205d9a50ff

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\ac48091f-a4f4-4268-b968-75205d9a50ff\subagents\workflows\wf_6abc9156-811\agent-a576dc9404b23b1fd.jsonl`  
- SHA-256: `5b02fce24524cc01bd1114fdf7dd1f79f5674a41941a2753f123cc452053ad52`  
- Source modified: `2026-06-22T01:58:00+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_6abc9156-811`  
- session_id: `ac48091f-a4f4-4268-b968-75205d9a50ff`
```

Transcript

1. user (2026-06-22T01:55:49.369Z)

You are scanning planning docs to surface ACCURATE, prioritized next steps for a co-founder.

REPO RULES:

```
- Engine repo at /c/Users/avidu/Projects/badminton-highlight-indexer: its working tree is on a  
STALE serving branch. Read DOCS + verify SHIPPED state from ENGINE origin/master ONLY:  
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer show origin/master:docs/FILE.md  
(read a doc on master; if it errors, the doc may be branch-local ? fall back to the working-tree  
file and SAY SO)  
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer log origin/master --oneline -60  
(what actually shipped ? use this to catch stale "TODO/next" claims)  
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "PATTERN" origin/master  
-- 'PATHGLOB'  
- khelsutra repo at /c/Users/avidu/Projects/khelsutra: on clean master ? use Read/Grep/Glob  
directly + git -C /c/Users/avidu/Projects/khelsutra log --oneline -40.
```

ANTI-SPIRAL (critical): before listing any "[gap]/net-new/TODO/next" item, VERIFY it hasn't already shipped ? cross-check the doc's claim against git log origin/master. A doc saying "X is next" is a HYPOTHESIS until you confirm X isn't already merged. Flag any stale doc claims you correct.

Context (already TRUE as of this session, do not re-list as "next"): the khelsutra ALPHA is DEPLOYED + live on a CPU droplet (engine + cloudflared tunnel + Cloudflare Access + SPA on Pages, real Gemini key working, full chain verified). DEPLOY_ALPHA fixes + the droplet kit are merged (khelsutra #78/#79/#81/#85). T2 (alpha deploy) is effectively DONE.

Return prioritized next items with honest status + git grounding.

TRACK: ****engine serving / platform / owned-model****. Read ENGINE origin/master docs: `docs/NEXT\_STEPS.md`, `docs/DEFERRED\_HARDENING\_PLAN.md` (esp. #16 async job model = the committed next PLATFORM slice), `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (Phase-0 milestone), and grep for the COMPUTE_DECOUPLED_SERVING / M-series state (M1?M12 ? which merged? `git -C /c/Users/avidu/Projects/badminton-highlight-indexer log origin/master --oneline -60 | grep -iE "M[0-9]|serving|compute"`).

ANSWER: what's the committed next serving/platform slice, what's owner-gated (owned-model Phase-0, M9-consent/ToS, M11 flywheel, M12 gdrive), and the UI-driven-cloud-serving gap status. Prioritize, marking owner-gated clearly.

2. assistant (2026-06-22T01:55:53.050Z)

I'll scan the engine planning docs and verify shipped state from origin/master. Let me start by pulling the latest and reading the key docs.

3. user (2026-06-22T01:55:57.203Z)

=== FETCH DONE ===

a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers (#309)
de2bf14 feat(pkg): P2e ? lazy-torch so the LIGHT tier imports with NO torch installed (default-OFF) (#308)
679e642 feat(pkg): P2d ? DB upgrade-safety (downgrade guard + backup-before-migrate + export) (default-OFF) (#307)
f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale (#306)
fa7ee3f feat(pkg): P2a ? single ffmpeg/ffprobe resolver (env-overridable; one fail-loud place) (default-OFF) (#305)
e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts) (#304)
ca34787 feat(pkg): P2b ? friendly `indexer --app` launcher (boot localhost server + open the bundled UI) (#303)
4a06b8b feat(pkg): P1b ? bundle the KhelSutra SPA single-origin into the engine wheel (default-OFF) (#302)
be0c434 feat(pkg): P1a ? net-new console/wizard endpoints: /api/version, /api/capabilities, /api/videos, /api/reels (#300)
cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state (default-OFF) (#298)
e8bad40 feat(pkg): P0d ? declare scipy (undeclared serving dep) + license-scan & built-wheel CI lanes (#297)
3895963 feat(pkg): P0c ? config/profile-driven bind address + DNS-rebinding Host-header guard (default-OFF) (#296)
fc78fce feat(pkg): P0b ? redact secret-shaped fields from the unauthenticated GET /api/config (#295)
0c0e064 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed engine (default-OFF) (#294)
82a28f4 feat(pkg): P0a ? resolved app-data HOME (RALLY_APP_HOME) for config/.env/DB/output (default-OFF) (#293)
2afbadf fix(dryrun): install the watermark font in bootstrap + track the UI-trigger asks (#292)
7835f34 feat(gcp): always-install Ops Agent on GPU VMs + a cloud-GPU dry-run guide (#291)
aac7633 feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF) (#290)
e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching (#289)
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF) (#288)
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked) (#287)
b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity test (default-OFF) (#286)
fef412 docs(serving): resolve §8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)
06ff4b7 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id (B-P1 guard) (#284)
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)
2afd54f feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF) (#282)
41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)
e111add feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity) (#280)
ce22791 feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-OFF) (#279)
f518e97 docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar, consent-not-surprise) (#278)
49c082a docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness (M11) (#277)
f592281 docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project (#275)
f6f15ca docs: isolate field-ops/PMF + Roora into archives/; carve a no-ML docs/data_pipeline/ (#276)
03fa951 docs: resolve DOC_STATUS findings #1 (per-video supersede) + #2 (NEXT_STEPS refresh) (#274)
aa5e9db docs: add DOC_STATUS register + archival due-diligence convention (#273)

bd42250 docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014 (#272)
4957b72 fix(worker): configure logging + explain a 0-segment infer (the cold-start blind spot) (#271)
4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)
c8e60ee fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball (#269)
e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)
a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted (\$) (#267)
b05401b docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013) (#266)
4d546ec fix(gcp): teardown defaults to DELETE (a stopped VM still bills its disk) (#265)
dd30e9b feat(workspace): per-video workspace convention ? design doc + Phase 0 helper (#264)
dc9111b docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is the serving model (#263)
d8c5aa0 docs(deploy): report polish ? highlight reel smoke-test result + VM-deleted final state (#262)
0d73d96 docs(deploy): first real GCP-L4 run report + gpu_setup conda-ToS/deps fix + Ops Agent runbook (#261)
380a4d0 docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; DO=train, GCP=serve (ADR-011) (#260)
36ed29f feat(wasb): native runner defaults to streaming + correct M3.5 parity gate to a tolerance (#259)
38c2d0f feat(storage): gdrive:// reads a locally MOUNTED Google Drive (no API) (#258)
55ae408 docs(decoupled): GCP pivot runbook + DO?GCP ADR-010 (deploy/gcp + handoff) (#257)
dc53db5 feat(detectors): Linux-native WASB runner (M3.5) + worker real-trajectory extraction (#256)
5c77f48 docs(decoupled): HANDOFF.md ? session checkpoint / resume-here (#255)
d5407eb feat(worker): Gen-0 train-from-bucket (worker train) on a CPU stub-trajectory mock (#254)
814d813 feat(deploy): GPU machine provisioning ? NVMe scratch + native WASB setup (#253)
f06c6a6 fix: cross-platform canonicalization (Linux/macOS, not just Windows) (#252)
6e5ac9e feat(worker): python -m backend.worker dispatcher ? bucket-driven train/infer (#251)
d292854 docs: SETUP_NEW_MACHINE ? canonical zero?train/infer runbook (#250)
6878d45 feat(storage): pluggable StorageBackend ? local + S3/GCS first-class buckets (#249)

4. user (2026-06-22T01:55:57.523Z)

f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale (#306)
cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state (default-OFF) (#298)
e8bad40 feat(pkg): P0d ? declare scipy (undeclared serving dep) + license-scan & built-wheel CI lanes (#297)
0c0e064 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed engine (default-OFF) (#294)
aac7633 feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF) (#290)
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF) (#288)
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked) (#287)
b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity test (default-OFF) (#286)
fefa412 docs(serving): resolve §8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)
2afd54f feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF) (#282)
41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)
e111add feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity) (#280)
ce22791 feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-OFF) (#279)
f518e97 docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar, consent-not-surprise) (#278)
49c082a docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness (M11) (#277)

bd42250 docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014 (#272)
 4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)
 e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)
 a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted (\$0) (#267)
 b05401b docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013) (#266)
 dc9111b docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is the serving model (#263)
 380a4d0 docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; DO=train, GCP=serve (ADR-011) (#260)
 36ed29f feat(wasb): native runner defaults to streaming + correct M3.5 parity gate to a tolerance (#259)
 55ae408 docs(decoupled): GCP pivot runbook + DO?GCP ADR-010 (deploy/gcp + handoff) (#257)
 dc53db5 feat(detectors): Linux-native WASB runner (M3.5) + worker real-trajectory extraction (#256)
 5c77f48 docs(decoupled): HANDOFF.md ? session checkpoint / resume-here (#255)

5. assistant (2026-06-22T01:56:00.585Z)

The git log is rich. Let me read the key planning docs from origin/master to verify status claims.

6. user (2026-06-22T01:56:01.747Z)

Next steps (live) ? refreshed 2026-06-21

> **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now **one of several active tracks**. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC_STATUS.md`.

? Where we are (2026-06-21)

DECOUPLED_COMPUTE delivered on GCP + archived (#270); **compute = GCP only** (ADR-013, DO dropped). **COMPUTE_DECOUPLED_SERVING** subproject spun up (#272, **ADR-014**) ? one pure core + local/cloud profiles; **owner M0** design sign-off pending. **Per-video workspace P0** shipped (#264); **doc-status** register + archival due-diligence in place (#273). The **khelsutra** site + **per-rally UI** ship in parallel (sibling track). The **rally-quality / owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.

? Prioritized pending ? all tracks

P0 ? highest leverage / unblocks the most

0. **owner-priority, 2026-06-21** MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH". Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12** does NOT fix it. Needs (a separate **khelsutra web/SPA** + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a **persistent** endpoint), **(ii)** the **SPA ingest/progress screen** (**khelsutra web/ B-P2: enter/upload video ? /api/process ? poll /api/jobs ? reel; ZERO engine code**), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER** in the UI ("your machine vs our cloud", D13/D15). The engine seams make it **possible**; the SPA makes it **usable**. Full record: `[ui-driven-cloud-serving-gap]`.

0b. **packaging track, IN PROGRESS** ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING_PLAN.md`. The **LIGHT-tier** engine is bundle-ready: `pip install ? indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **Next (fresh focused pushes):** **(1)** P3 **GPU validation + screencast** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default config), **(3)** P2c first-run wizard (cross-repo **khelsutra SPA** from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

1. ? **COMPUTE_DECOUPLED_SERVING** M0?M9 LARGELY SHIPPED (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5** #286, M6 #287, M8 #288, M9-auth #290 all merged (default-OFF); the

****real M7 L4 run**** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). ****Owner residual**** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a ****\$100/?10k cap**** (D17).

2. ****[owner, ongoing] Grow the golden corpus**** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track ***and*** full-corpus training feed on it.

3. ****[owner + Claude · GPU] Full-corpus training (n?5)**** ? a real ``weights/<ver>/`` (the ``0.1.0-l4`` bundle is n=2 plumbing-only).

****P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)****

4. ****[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12**** (M5/M6/M8/M9-auth ? merged this session). ****M11**** = ``backend/flywheel.py`` capture?store?shadow-eval?goldenize plumbing reusing ``workspace.py::for_video`` + ``backend/eval/experiment.compare`` + ``backend/eval/promotion.py::promote_if_served_safe``, behind ``flywheel.*`=OFF` + a faked ``consent_attested`` (? privacy-sensitive ? give it fresh focus). **\*\*M12\*\*** = ``backend/storage/gdrive_api.py`` ``GDriveApiStorage(StorageBackend)`` + a ``gdrive-api://`` scheme (regex already allows the hyphen; add to ``_KNOWN`` + ``_backend_class`` + the `StorageConfig Literal`) against an INJECTED fake Drive (mirror ``gcs.py`` DI); keep the mount-only ``gdrive://`` untouched. ? ****M9-consent cols = a new v7 migration**** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.

5. ****[Claude] Per-video workspace P1?P6**** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of ``detector/``+``tmp/`` *[the ``output/chunks_`` de-hardcode base = ? done via #282]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.

6. ? ****Ops-Agent codified (#291)**** ? ``deploy/gcp/create_gpu_vm.sh`` (the one-command VM helper) always bakes the Ops-Agent startup-script; `provision.sh` grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])

7. ****[owner] Set the ship-gate target (#172)**** ? needs corpus growth.

8. ****[owner-side GPU] R-series: promote-or-reject `R2` to F1 as the gate**** (the ablation) ? needs GPU-extracted golden features.

****P2 ? later / gated / deferred****

9. Rally-quality ****P3/P4**** (person-fusion LOVO, audio cue) ? needs GPU golden features.

10. ****M0.4 scorer swap**** ? TemporalMaxer (per the 2026-06-17 external review).

11. ****Multisport**** (table-tennis first, then pickleball).

12. Serving ****M8/M9**** ? per-video cost ledger, edge auth, DPA/consent (within ``COMPUTE_DECOUPLED_SERVING``; owner/legal-gated).

13. ****WASB-C3**** weight provenance (gates ***sharing*** a trained model) ? legal.

14. ****Rename repo off "badminton" (owner, BACKLOG).**

15. Owner housekeeping: retire the shared ****DO droplet**** + ****DO token****; collector ****`md5`**** export (collector repo).

16. ****PMF field validation**** (owner-side, business track).

****Doc-hygiene (tracked in `docs/DOC\_STATUS.md` \$4):**** ? #1 per-video reconcile + ? #2 this refresh; ****pending:**** field/PMF/Roora ? archives + a data-layer ``docs/`` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify ``HOSTING_PLAN`/`UI_PROPOSAL`` currency.

Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)

****Updated:**** 2026-06-13 (****rally-detection quality track****: fusion + experiment-harness design docs merged,

owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the ****UI-alpha track opened****:

ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private ``khelsutra`` repo, local-first delivery ? see

``docs/UI_ALPHA_EXECUTION_PLAN.md``; ****PR-1 async jobs (#16) MERGED 2026-06-11, PR #87****; ****PR-2 upload/download MERGED 2026-06-11, PR #99**** ? live-E2E-verified (3-part chunked upload MD5 byte-identity +

compile?browser-download); next: PR-3 identity. Same day: the ****8?9 quality sweep #90?#98 landed**** ? ruff/mypy/

coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into
`trajectory\_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
****Authoritative roadmap:****
`docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (+ ADR-008 for topology). ****Latest blow-by-blow:****
memory
`current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.

? Next-session / GPU-box pickup (2026-06-15)

****M0 in-container quick wins are COMPLETE:**** ship-gate provisional blocker (****#170****), train/eval split guard (****#171****), the ****doc-consistency sweep**** (****#174**** ? corrected 16 drifted docs vs the code), and the ****P3 person-feature fusion litmus machinery** is already present + suite-green (``backend/eval/fusion_compare.py`` + ``ablation.py``: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden ``skipif`` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a ****GitHub Actions minutes/billing outage**** (account-level, not code) ? work is local-gated + ``--admin``-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

****Immediate next step ? the real P3 LOVO measurement**** (this box has a GPU + ``torch+cu124`` + ``cv2``; videos and trajectory CSVs are present, but the GPU-extracted golden features + ``rallies.csv`` labels are not yet here):

- **[owner / GPU box]**** ``python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6`" ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `rallies.csv` labels are owner-side only.)`
- Artifacts land in the ****shared cloud folder**** ``C:\Users\avidu\OneDrive\rally-annotator\golden-data\`` (OneDrive auto-syncs both ways). Convention + workflow:
****[`GOLDEN\_DATA\_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)****; tracked in ****#175****.
- **[CPU dev/agent session]**** ``python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record`` + ``python -m backend.eval.ablation --features-dir <shared>\... --granularity group`` ? real person-feature ?F1; copy the JSONs into ``output/`` to also pass the ``skipif`` litmus tests (``tests/test_ablation.py::test_litmus_reproduces_gate_a``).

****Also queued (owner-gated / pending input):****

- Ship-gate target calibration ? ****#172**** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

****Leaving (bulky / M2 ? no start without owner go):**** ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

Where we are (one paragraph)

ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (``training/``) behind a CI-enforced import wall; the featurizer is single-sourced (``backend/features/rally_seq.py``, `train==serve`); the ****Gen-0 harness ships**** (``python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor eval_baselines/heuristic_lovo_n6.json --version <semver>``) and produced artifact v0.1.0 (LOVO ****0.626****, floor passed, sign test 5/6 wins $p=0.109$? honestly not significant, ship=False). The ****n=6 owned corpus** is in the collector (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The ****Gemini battery is done**** (see table) ? the moat is quantified.

The quality table that now drives strategy (IoU 0.5 vs human golden labels)

Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
--- --- ---		
Heuristic (velocity windowing)	0.657	0.157

```
| **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |  
| Two-stage WASB+Gemini refine | 0.83 | ? |  
| **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
```

Reading: clean footage is commoditized (serve it with Gemini for cents). **Multi-court drops the commodity**

wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges,

the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an

architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes

(uploads candidate clips

only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.

Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)

This is the track a desktop/local agent is slated to pick up. Design docs landed (PR #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR off `master`.

Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs

#156/#157/#160/#161/#163 + #165 + final records). See archived

`docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every

pre/post/review) and `docs/archives/HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive

done; clean slate.

- **Docs:** `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (shuttle + person + audio, one fusion layer) .
`docs/EXPERIMENT\_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
`docs/ROORA\_LABELING\_GUIDELINES.md` (v8) . `docs/HOW\_RALLY\_DETECTION\_WORKS.md` (2-layer mental model).

- **Key finding** (no re-investigation needed): the owner's "held-shuttle counted as a rally" bug is **NOT** a missing capability ? `backend/pipeline/detectors/rally\_gate.py` already implements the net-crossing **"Gate A"** (`apply\_gate(..., min\_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only** called from eval drivers** (and there was no `min\_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter** (now via fusion_hybrid) was the single cheapest, highest-confidence quality win. (P2 FusionSegmenter specifics: registry `fusion\_hybrid`, **default-OFF**, seeds from substrate, persists `net\_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test\_served\_gate\_a\_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)

- **Current status** (2026-06-14): P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next** (this PR + follow-ups): learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.

- **Pick-up order** (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):

1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)

2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion\_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).

3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

- **Discipline** (from EXPERIMENT_HARNESS.md): every new cue **flag-gated**, default OFF; promote to default **only** on measured LOVO lift through `run\_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

- ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `rallies.csv` labels, which are not yet present in this checkout.

- **External due-diligence review inputs** (2026-06-17) ? PICK UP AFTER current ideas are

exhausted; full details + citations in RALLY_QUALITY_RESEARCH.md

§5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):

1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as "the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."
2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.
3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).
4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.
6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

Discipline reminder: owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

Prioritized next steps

1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
Priority order for new videos: *(a) multi-court badminton* (the target distribution AND where the ship target lives), *(b) ?3 matches per new sport ? table tennis first* (WASB transfers at F1 0.909; pickleball labels are corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per VIDEO_RECORDING_GUIDELINES.md; **adult sessions only for the training pool** (consent guardrail). Per video:
label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.
The paired sign test needs n: at n=10 with 9 wins, $p > 0.011$? significance is reachable this month.
2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with paired-sign $p < 0.05$ vs heuristic" ? owner to ratify/adjust.
3. **M0.4 next increment (\$0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review (`RALLY\_QUALITY\_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated by C3 (provenance multiplies per sport).
5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
"multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:
`output/MahadevpuraSingles\_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified

the
 physical-visit approach; reframed product-feedback-first per owner spec):**
 `docs/PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md` (the recruiting flyer ? owner fills ?/contact
 placeholders and recruits) +
 `docs/FIELD\_VISIT\_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script,
 demo-after-Q6
 rule, ground rules, Phase-2 record-and-show-back stretch goal) +
 `docs/FIELD\_VISIT\_ANSWER\_SHEET.md` (per-visit
 capture form incl. demo-reaction + student blocks) + `docs/PMF\_FIELD\_SIGNALS.md` (G/A/R
 anchors + **pre-registered
 decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
 6. **Housekeeping:** ~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working;
 confirm old-key
 deletion in AI Studio); **rename repo off "badminton"** (owner action,
 in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery
 completed.

Pointers / artifacts

- Roadmap: `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` · Topology: ADR-008
 (`docs/archives/decisions/DECISIONS.md`) ·
 Study: `docs/archives/research/OWNED\_MODEL\_TRAINING\_STUDY.md` · Quality history:
 `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
- Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle;
 weights gitignored,
 manifest reviewable). Featurizer: `backend/features/rally\_seq.py` (FEATURE_SCHEMA v1).
- Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
 `output/human\_lovo\_manifest.json`
 (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
 Setup\Collect\Trajectories\` ·
 labels: `...\Collect\Golden Labelled\`.
- Gemini ops learnings: serial uploads (`indexing.ai\_max\_workers=1`) + chunk re-encode
 (`indexing.ai\_chunk\_scale\_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid
 project burst-throttles
 with a MISLEADING "credits depleted" 429; ramp gradually.
- ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
 milestone. Committed next
 PLATFORM slice = async job model (single-tenant).

7. user (2026-06-22T01:56:03.346Z)

040000 tree 4eb0c71d2de6992f9f1cd2934aa42b22d861273f	docs/COMPUTE_DECOUPLED_SERVING
100644 blob 3ce9e584a066576a89949868752f3d82524e4f2d	docs/DEFERRED_HARDENING_PLAN.md
100644 blob 875ab15de7ceaa2501034126521ff6ecabad9ad4	docs/NEXT_STEPS.md
100644 blob 04d84eb21190ef54cd433cedf2c25771ec746326	docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md

8. assistant (2026-06-22T01:56:07.195Z)

NEXT_STEPS.md is rich and recent (2026-06-21). There's a `docs/COMPUTE\_DECOUPLED\_SERVING/`
 directory (subproject). Let me read the subproject tracker and the deferred hardening #16, plus
 check what M-series state is documented vs shipped.

9. user (2026-06-22T01:56:08.795Z)

100644 blob ec9f84b38afe1d1045a0f851effdb807054dae63	docs/COMPUTE_DECOUPLED_SERVING/README.md
100644 blob 373148f834520eb33cb8144b011d199b3b4a7972	
docs/COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md	
100644 blob 6a22393ef7df636770eb0682d06b744922290b9d	
docs/COMPUTE_DECOUPLED_SERVING/architecture.svg	
040000 tree ff2d7f636c3e0baebbb722f6d40f5e4e83cd4812	docs/COMPUTE_DECOUPLED_SERVING/runlogs

10. user (2026-06-22T01:56:10.428Z)

Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging (COMPLETE

****STATUS: COMPLETED**** (2026-06-14). This planning doc's three items are all dispatched:

- ****#16 Async job model ? DONE**** (PR #87; hardened by the self-scheduling drain loop in #149).
- ****#7 Packaging modernization ? DONE**** (PR #167: PEP 621 `pyproject.toml`, `setup.py` ? `scripts/doctor.py`, `indexer` console entry point, CI switched to `pip install -e .[dev]`).
- ****#13 Train/eval separation guardrails ? moved to BACKLOG.md**** (?; the concrete `splits.py` / `assert\_disjoint` match-level-split design is preserved there).

Full archived snapshot (the original plan + per-item design/risks/verification):

[`archives/DEFERRED\_HARDENING\_PLAN-COMPLETED-2026-06-14.md`](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md).

_(This top-level file is a pointer stub kept so existing links to
`docs/DEFERRED\_HARDENING\_PLAN.md` still resolve; the content now lives in the archive.)_

=== #16 SECTION ===

1:# Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging (COMPLETED)

5:- ****#16 Async job model ? DONE**** (PR #87; hardened by the self-scheduling drain loop in #149).

11. assistant (2026-06-22T01:56:14.830Z)

Important correction: DEFERRED_HARDENING_PLAN #16 (async job model) is ****DONE**** (PR #87), not "the committed next slice" ? the task framing called it the committed next slice, but the doc says it shipped. The committed next PLATFORM slice has moved on to the COMPUTE_DECOUPLED_SERVING subproject. Let me read that subproject README (the M-series tracker) and the OWNED_MODEL plan Phase-0.

12. user (2026-06-22T01:56:15.651Z)

<persisted-output>

Output too large (36.2KB). Full output saved to: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\ac48091f-a4f4-4268-b968-75205d9a50ff\tool-results\b7hqk0l7g.txt

Preview (first 2KB):

Compute-Decoupled Rally Serving ? one pure core, profile-selected compute + storage

```
> **Status:** ? **M0 APPROVED** (owner, 2026-06-21) ? **IN PROGRESS (build M1+)** . **Owner:**
Avi . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/past_projects/`
when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory `current-
state` + `docs/NEXT_STEPS.md` once work starts.
> **Relation to existing docs:** the productionization successor to the archived
`DECOUPLED_COMPUTE` (M3/M4); realizes the `ComputeBackend`/`StorageBackend` registries from
[PLATFORM_ARCHITECTURE.md](../PLATFORM_ARCHITECTURE.md) §3; the **truly-local** profile is the
L0 tier of [VIDEO_LOCALITY_MODEL.md](../VIDEO_LOCALITY_MODEL.md) and the `LocalDesktop` backend
of [DESKTOP_APP_PLAN.md](../DESKTOP_APP_PLAN.md).
> **Naming:** this was the "Cloud Run + GPU serving subproject" (ADR-012). Renamed **compute-
decoupled serving** because **truly-local is a co-equal, first-class profile**, not an
afterthought.
> **Honesty note:** claims tagged `[verified]` (checked against code by the design workflow
`wf_56274ff0`), `[design]` (proposed), `[researched]`.
```

```
> **? Northstar (D14):** a **mobile, app-like** flow ? point at a **Google Drive** video ?
**fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next
to the source**. Two design promises around it: a user can **switch local?cloud anytime** (D13,
a headline feature), and execution is **never a surprise** ? auto-detect **suggests**, the user
**confirms** (D15).
```

0. TL;DR

...

</persisted-output>

13. user (2026-06-22T01:56:20.872Z)

=== PROGRESS TRACKER (\$7) ===

7. Deliverables & progress tracker ? **source of truth**

Legend: ? Todo · ? In progress · ? Done · ? Gated. **One small PR per row**, off
`origin/master`, no stacking. Default-OFF where it touches core callers.

```
| ID | Deliverable | Depends | Gated? | Status | PR |
|----|-----|-----|-----|-----|----|
| M0 | This design doc + project dir + `runlogs/` + ADR-014 | ? | owner sign-off | ? | **?
APPROVED 2026-06-21** |
| M1 | `deployment.profile` + `compute_target` config + preset bundles + precedence/auto-detect;
unit tests (no behaviour change, profile='' default) | M0 | safe | ? |
[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279) |
| M2 | `input_backend`/`input_root`; wire main CLI/API input through `StorageRef.parse_uri`
(local=identity); storage-fake tests | M1 | safe | ? |
[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280) |
| M3 | Configurable output/scratch base ? kill hardcoded `output/` (`trajectory_hybrid:237,313`,
`yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**; golden + stub-E2E byte-identical | M1
| ? default-OFF + Launch Report on flip | ? | [#282](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/282) |
| M4 | `DeviceContext` + WASB **CPU-fallback**; unified healthcheck/device wiring | M1 | safe
(cuda default unchanged) | ? | [#283](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/283) |
| M5 | **truly-local profile** end-to-end (preset + startup checks); first committed **runlog**
(truly-local sample run) | M2,M3,M4 | owner real-GPU | ? |
[#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-
checks+L4 parity ?; runlog ? owner GPU) |
| M6 | Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) +
**containerized worker image** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner
(GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)
(shim+image+mocked ?; real build/run = M7 ? owner) |
| M7 | **cloud-serving profile** e2e on L4: upload<2GB` + gdrive/bucket ? detect+stitch on
Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet) | M6
| ? owner (spend) | ? | ? |
| M8 | Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds, wall_seconds,
bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation + `/api/.../cost` | M7
| ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission ?
owner/follow-up) |
| M9 | **Edge auth** (Cf-Access extraction) + per-tenant scoping on `user_id`; DPA/consent gate
wiring | M7 | ? owner (privacy/counsel) | ? | [#290](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/290) (auth: Cf-Access JWT verify + owner_id tag, default-OFF ?; **+
exposure-safety boot posture** in `startup.py` ? edge-auth-on requires the `allowed_video_root`
jail + CF team/AUD or the server refuses to start, see changelog; consent cols + ToS/DPA ?
owner/counsel) |
| M10 | `byo_worker` / zero-infra-BYO ? **design-only, DEFERRED** | M8,M9 | ? explicit owner
approval | ? | ? |
| **M11** | **Data-flywheel harness (D12):** on a consented adult job, **capture** the upload +
Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-eval**
owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the golden
TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the owned model
climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent gate (M9). |
M7,M9 | ? owner (consent/privacy) | ? | ? |
| **M12** | **`gdrive-api` (OAuth) StorageBackend (D14 Northstar):** cloud reads the source from
+ writes the stitched reel **back to the user's Google Drive next to the source** (per-user
OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local mount
backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? | ? |
```

****Testing strategy is a first-class deliverable** ?**

```
[`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). **Run logs / sample runs for posterity** ?
[`runlogs/`](runlogs/).
```

8. Open questions ? owner *(blocking-gates vs nice-to-knows)*

14. assistant (2026-06-22T01:56:24.767Z)

Now I have the complete M-series state. Let me verify the M11/M12 claims against actual code on origin/master (do they exist or not yet?) and check the UI-driven cloud-serving gap status. Let me also verify M5/M6/M8/M9 code presence to confirm the "merged" claims.

15. user (2026-06-22T01:56:30.064Z)

```
=== flywheel.py (M11) present? ===
=== gdrive_api / gdrive-api (M12) present? ===
100644 blob bf51d7da3f747f56a7912fb1eda61cdb58e57470 backend/storage/gdrive.py
=== grep gdrive-api scheme ===
=== M9 consent cols / v7 in DB ===
origin/master:backend/infrastructure/database.py:82: ``user_version`` stays un-bumped ?
and the next open would re-run the same
origin/master:backend/infrastructure/database.py:101: # Versioned migrations live
here. PRAGMA user_version is the schema
origin/master:backend/infrastructure/database.py:104: version =
cursor.execute("PRAGMA user_version").fetchone()[0]
origin/master:backend/infrastructure/database.py:140: # future: if version < 8:
cursor.execute("ALTER TABLE ..."); PRAGMA user_version = 8
origin/master:backend/infrastructure/database.py:240: cursor.execute("PRAGMA user_version
= 1")
origin/master:backend/infrastructure/database.py:267: cursor.execute("PRAGMA user_version
= 2")
origin/master:backend/infrastructure/database.py:279: cursor.execute("PRAGMA user_version
= 3")
origin/master:backend/infrastructure/database.py:299: cursor.execute("PRAGMA user_version
= 4")
origin/master:backend/infrastructure/database.py:337: CREATE UNIQUE INDEX IF NOT
EXISTS idx_user_models_user_version
origin/master:backend/infrastructure/database.py:347: cursor.execute("PRAGMA user_version
= 5")
origin/master:backend/infrastructure/database.py:394: cursor.execute("PRAGMA user_version
= 6")
origin/master:backend/infrastructure/database.py:403: cursor.execute("PRAGMA user_version
= 7")
```

16. user (2026-06-22T01:56:32.802Z)

```
=== M5/M6 worker + cloudrun files ===
100644 blob 50642edae485b53214cda2553b90fd7970700493 backend/api/job_worker.py
100644 blob 86ff71e4a0cadb206f4e1d62c56d56b661b67803 backend/compute/cloud_run.py
100644 blob f8c4e114435fd6152dad5fa573063b9ff42e172 backend/config/startup.py
100644 blob 5ec992ea2b13064ea1e8580d7010419a7f4b5964 backend/pipeline/detectors/device.py
100644 blob 2f27c9ed72ba6037990487580fb4ac5d18d89fc6 backend/worker/__init__.py
100644 blob 081d8f58fc15f3b1c6c7b2703466b15ff58020af backend/worker/__main__.py
=== latest DB migration version ===
origin/master:backend/infrastructure/database.py:136: self._migrate_to_v6(cursor)
origin/master:backend/infrastructure/database.py:139: self._migrate_to_v7(cursor)
origin/master:backend/infrastructure/database.py:140: # future: if version < 8:
cursor.execute("ALTER TABLE ..."); PRAGMA user_version = 8
origin/master:backend/infrastructure/database.py:240: cursor.execute("PRAGMA user_version
= 1")
origin/master:backend/infrastructure/database.py:267: cursor.execute("PRAGMA user_version
= 2")
origin/master:backend/infrastructure/database.py:279: cursor.execute("PRAGMA user_version
= 3")
origin/master:backend/infrastructure/database.py:299: cursor.execute("PRAGMA user_version
= 4")
origin/master:backend/infrastructure/database.py:347: cursor.execute("PRAGMA user_version
= 5")
```

```

origin/master:backend/infrastructure/database.py:349: def _migrate_to_v6(self, cursor):
origin/master:backend/infrastructure/database.py:350:     # v6: per-job COST LEDGER
origin/master:backend/infrastructure/database.py:394:     cursor.execute("PRAGMA user_version
= 6")
origin/master:backend/infrastructure/database.py:396: def _migrate_to_v7(self, cursor):
origin/master:backend/infrastructure/database.py:397:     # v7: record the UI LOCALE a job
was submitted in (the language the SPA was rendered in,
origin/master:backend/infrastructure/database.py:399:     # clients / pre-v7 rows). Enables
PMF analytics ("which language markets use the tool",
origin/master:backend/infrastructure/database.py:403:     cursor.execute("PRAGMA user_version
= 7")
origin/master:backend/infrastructure/database.py:632:     ``locale`` (v7) is the UI language
the submission came from ? NULL when unrecorded.""
origin/master:backend/infrastructure/database.py:644:     CLI / pre-v7 / older clients)
bucket under '(unset)'. Returns {locale: count}, busiest first.""

```

17. assistant (2026-06-22T01:56:38.993Z)

Confirmed:

- **M11** (flywheel.py) ? NOT shipped. No `backend/flywheel.py` on origin/master.
- **M12** (gdrive-api OAuth StorageBackend) ? NOT shipped. Only the mount-only `gdrive.py` exists; no `gdrive-api` scheme.
- DB is at **v7** (M8 took v6, locale took v7). So the M11-consent migration would need **v8** ? the NEXT_STEPS note saying "M9-consent cols = a new v7 migration (M8 took v6)" is now **stale**: v7 was taken by locale (#306), so M9-consent/M11 needs **v8**.

Let me verify the OWNED_MODEL Phase-0 status and the M7 "real L4 run validated" claim, plus check the khelsutra side for the SPA ingest screen (B-P2) status.

18. user (2026-06-22T01:56:41.766Z)

=== STATUS HEADER ===

Owned-model implementation plan (roadmap) ? DRAFT for owner review

Status: PLAN / roadmap. No owned-model *implementation* starts until the owner greenlights a specific milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a training source).

Grounded in: `docs/archives/research/OWNED\_MODEL\_TRAINING\_STUDY.md`, the n=6 LOVO + learned prototype (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy workflow (2026-06-09).

Design principle (owner directive): **envision now, build later**

Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be **architecturally envisioned now** so we never hit a mammoth refactor. We **design the seams** (data contracts, export boundaries, the event store) up front and **implement behind them** when each milestone is greenlit. Punting a **feature** is fine; punting its **seam** is not.

North Star (reframed per PR #61 review)

- **Immediate deliverable:** a **training framework + harness** that produces a model with **very high precision/ recall for extracting highlights + rallies from a clip**. **Quality is the first-order objective.**
- **Eventual product:** a **single, sport-agnostic, conversational** video model ? upload a clip, then ask contextual questions ("longest rally", "make a clip of all service faults", "rallies over 20 shots").
- **On-device is LATER** (a model-compression goal so power users save upload cost/hassle) ?

definitely on the
roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
- **The moat = the consented dataset + the eval harness, not the weights.**

Hard guardrails (non-negotiable ? "never corrupted")

- **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT / Apache-2.0 / BSD only.**
(Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-restrictive: Gemma 3 (viral),
Gemma 3n (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU cap), Commons-Clause repos.
- **Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.** Also forbidden: public grounding-CoT datasets that
were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision (a real, binding cost).
- **Exclude minors; per-user training data needs a separate explicit opt-in; split data by match.**
- ? **Base-model pretraining provenance is unauditably for every open VLM (incl. Qwen3-VL).**
Our "no paid-LLM
training" rule is enforceable at **our fine-tuning data** layer, not the base's pretraining ? an **explicit owner**
risk-acceptance is required to use any open VLM. (Governed by the base's license, not our pipeline.)

The model strategy (verified 2026-06-09)

Framework ? base (the Gemma-4-vs-ms-swift clarification): a **framework** is the forge (runs SFT/GRPO, emits weights); a **base** is the metal you forge. You pick one framework and feed it any clean base.
- **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio + timestamp-grounding + GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core; **avoid** its AGPL Studio UI*)
as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-time snapshot ? re-verify at impl.
- **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-video context, purpose-built
=== PHASE 0 ===
3:**Status:** PLAN / roadmap. No owned-model **implementation** starts until the owner greenlights a specific
4:milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a training source).
11:event store) up front and **implement behind them** when each milestone is greenlit. Punting a **feature** is fine;
99:## Phase 0 ? Foundation (start now, per-milestone greenlight; \$0, local, no cloud/DPA)
101:### M0.1 ? The labeled-corpus spine **+** the event-index contract (the moat) . **greenlight item**
114:### M0.3 ? Compliance cleanup (**hard blocker ? do before any training run**) . **greenlight item**
125: Carry as a tracked blocker that **gates SHIP** (not Gen-0 research).
127:### M0.4 ? Gen-0 TAL harness (**FIRST-CLASS ACTION ITEM** per owner) . **greenlight item**
128:Productionize `rally\_seq\_proto.py` ? a **one-command train?eval?ship-gate** harness over **ActionFormer (MIT, 1:1**
130:reusing `metrics.match\_intervals`. **Define the ship-gate target up front** (open item): a FLOOR of "beat the honest n=6
136:## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
137:Run the A/B; if the learned Gen-0 clears the ship-gate with per-video calibration, it becomes the rally/highlight
140:## Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + healthy corpus + explicit go)
146:- **Hard pre-serving gates (unmeasured, load-bearing):** the **\$/call crossover** (self-hosted VLM vs Gemini) **+** a real
155:| 1 | Greenlight Phase-0 milestone(s) ? rec: **M0.1 corpus+event-index**, then **M0.4 harness** | awaiting owner |
162:## Open risks / unknowns (none block Phase-0 research)

163:Budget/cost to reach Gen-2 ****unquantified**** (largest risk) · ship-gate "very high P/R"
target undefined · per-sport clean
169:*Design-only and reversible. Nothing starts until you greenlight a milestone. Sequencing
discipline holds: the committed

19. user (2026-06-22T01:56:43.480Z)

17242a9 feat(site): localize own-model body content to kn/hi (#86)
7b04f02 deploy(kit): align with the live-verified droplet deploy (+ the gemini-default fix)
(#85)
5e5a2ac docs(friendly-multilingual): close out the chapter (§9 DoD + UI/UX + B7 deferred) (#84)
4c0f35c fix(site): UI/UX ? compact language switcher + header overflow fix + localized hero
(#83)
0fae364 feat(web+site): A2 ? localized WhatsApp share for the reel (SPA + site) (#82)
1e52cfd deploy(droplet): tailored alpha kit for the CPU droplet (engine api. side) (#81)
64076a5 test(web): add a coverage floor (CI-gated) + i18n regression tests for the #77 audit
(#80)
8328721 docs(tracker): refresh the §7 milestone prose to merged reality (A6/A7/#77 landed) (#79)
3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook
(RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
9f1ec9c fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)
9a2d7d2 feat(site): A7 ? camera-placement diagram on the capture page (#76)
aa8d5e9 feat(site): A6 ? plain-language Request-Access GPU/storage questions (#75)
1cd84af feat(site): A3 ? localize recording-education + marketing to kn/hi, Noto on the site
(#74)
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process
(analytics) (#73)
1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-
HOME) (#70)
f7e2bd5 chore(site): add robots.txt + a "crawling / bot protection" runbook for khelsutra.guru
(#71)
bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling
tracked (#301) (#69)
46e0244 feat(web): A9 ? localized tap-only filter controls (#68)
a2292ef feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs (#67)
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.*
tunnel + CF Access) (#65)
8bbc557 feat(web): VITE_API_BASE ? let the SPA target a split api.* origin (default same-origin
/api) (#64)
92a6c45 feat(web): A8 ? background-processing reassurance + icon+text status chips (#63)
501c014 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*)
(#62)
95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)
=== B-P2 / ingest / process screen in khelsutra ===
0fae364 feat(web+site): A2 ? localized WhatsApp share for the reel (SPA + site) (#82)
9f1ec9c fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process
(analytics) (#73)
a2292ef feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs (#67)
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
8bbc557 feat(web): VITE_API_BASE ? let the SPA target a split api.* origin (default same-origin
/api) (#64)
92a6c45 feat(web): A8 ? background-processing reassurance + icon+text status chips (#63)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280)
(#55)
0021c4f test(web): guard the SPA stays self-contained / offline-renderable (#50)
631c3aa docs(friendly-multilingual): SPA fully trilingual ? mark B1/B2/B4/B5/B6 done +
screencast milestone (#49)

20. assistant (2026-06-22T01:56:49.215Z)

Key findings confirmed. The khelsutra B-P2 (in-UI bucket ingest) shipped (#56), and T4 (in-UI
file upload) shipped (#66) ? so the SPA ingest/progress screen described in NEXT_STEPS P0 item

#0(ii) is actually ****already built****. This is a stale claim in NEXT_STEPS. Let me verify the SPA has the upload?process?poll?reel flow and check whether a compute picker exists (D13/D15), and confirm M7 status.

21. user (2026-06-22T01:56:55.799Z)

```
=== web SPA files ===
README.md
dist
index.html
node_modules
package-lock.json
package.json
src
tsconfig.json
vite.config.ts
=== grep for jobs polling + process + compute picker in web ===
web/dist/assets/index-Dt-eK0X5.css
web/dist/assets/index-DuWdxnSi.js
web/node_modules/.vite/deps/i18next-icu.js
web/node_modules/.vite/deps/i18next-icu.js.map
web/node_modules/.vite/deps/react-dom_client.js
web/node_modules/.vite/deps/react-dom_client.js.map
web/node_modules/@asamuzakjp/css-color/dist/esm/js/color.js
web/node_modules/@asamuzakjp/css-color/dist/esm/js/color.js.map
web/node_modules/@asamuzakjp/css-color/dist/esm/js/constant.d.ts
web/node_modules/@asamuzakjp/css-color/dist/esm/js/constant.js
web/node_modules/@asamuzakjp/css-color/dist/esm/js/constant.js.map
web/node_modules/@asamuzakjp/css-color/dist/esm/js/convert.js.map
web/node_modules/@asamuzakjp/css-color/dist/esm/js/relative-color.d.ts
web/node_modules/@asamuzakjp/css-color/dist/esm/js/relative-color.js
web/node_modules/@asamuzakjp/css-color/dist/esm/js/relative-color.js.map
web/node_modules/@asamuzakjp/css-color/dist/esm/js/resolve.js
web/node_modules/@asamuzakjp/css-color/dist/esm/js/resolve.js.map
web/node_modules/@asamuzakjp/css-color/dist/esm/js/typedef.d.ts
web/node_modules/@asamuzakjp/css-color/README.md
web/node_modules/@asamuzakjp/css-color/src/js/color.ts
```

22. user (2026-06-22T01:56:56.180Z)

```
=== M7 mentions in CDS runlogs / DEPLOY_REPORT ===
100644 blob bae22991a2cb04c19f52ad70c8ced92b47b83d54
docs/COMPUTE_DECOUPLED_SERVING/runlogs/README.md
100644 blob 84c080aae43cebc33ecf7838c34a2fc0b9af30c3
docs/COMPUTE_DECOUPLED_SERVING/runlogs/RUNLOG_truly-local_TEMPLATE.md
=== grep M7 status in CDS README ===
34:| D7 | **Cloud Run cold-start: SCALE-TO-ZERO** ? accept the ~1?3 min cold-start (it's an
async job: upload?poll?reel, amortized over the multi-minute job). **No warm pool** (that
~$500/mo idle would break per-video billing). | owner 2026-06-21 | M6/M7. Revisit a warm lane
only on a UX complaint. |
38:| D11 | **Quality floor: Gemini handoff ON** for cloud-serving until the owned model clears
the #172 ship-gate (e.g. ?0.70 multi-court) ? then flip via a **Launch Report**; the owned model
runs in **shadow/eval** meanwhile. (Gemini = serving/inference ONLY, never a training source ?
CLAUDE.md guardrail.) | owner 2026-06-21 | M7 + D12. |
43:| D16 | **M6/M7 implementation SHAPE + the cheap $8 picks (owner 2026-06-21):** (a) **Cloud
Run JOBS** (not Services-with-GPU) for the detect+stitch unit, **bucket-poll for completion**
(reuses `JobWaiting`/\`claim_due_job` verbatim, fits async upload?poll?reel + scale-to-zero D7);
(b) **stitch ENCODE = libx264** (protects the D5 byte-determinism/parity guarantee + the L4
parity test) ? **NVDEC for DECODE only** (a speed lever that never touches output bytes);
**NVENC encode is DEFERRED behind a config knob**, revisit only if M7 cost data justifies
relaxing parity ($8 Q8); (c) **input cap = hard-reject`<2GB`** for v1 (a config cap so raising
to 4GB later is one line; matches the shipped M2 chunked-upload threshold ? $8 Q9). | owner
2026-06-21 | M6/M7. Resolves $8 Q2 (co-located, = D4), Q8, Q9, Q10 (dir name confirmed). |
44:| D17 | **Real cloud verification BUDGET = $100 / ?10k cap (owner 2026-06-21):** the owner
authorized actual GCP spend for due-diligence + production-grade testing/verification of M5?M7
```


(real L4 run, the M5 runlog, the M7 DEPLOY_REPORT), ****provided the cap is respected**** ? confirm exact cmd + hourly cost before each spend, prefer the smallest viable run + cheapest-adequate GPU, and ****DELETE every resource after**** ([[gcp-vm-always-delete]]). Removes the M6/M7 "spend" gate within the envelope; the calibrated pricebook (M8) + counsel ToS (M9) + OAuth app (M12) stay owner/counsel-side. | owner 2026-06-21 | Unblocks the real M5/M7 runs (not just the code) within the cap. |

62:| DEPLOY_REPORT_GCP_2026-06-20 (archived) | First real L4 run; ****cross-GPU sub-pixel parity finding**** | Why D5 (parity at window level, not CSV bytes). |

120:| M5 | ****truly-local profile**** end-to-end (preset + startup checks); first committed ****runlog**** (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? |

[#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-checks+L4 parity ?; runlog ? owner GPU) |

121:| M6 | Cloud Run GPU ****dispatch shim**** (control plane ? job via storage ? re-claim) + ****containerized worker image**** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner (GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287) (shim+image+mocked ?; real build/run = M7 ? owner) |

122:| M7 | ****cloud-serving profile**** e2e on L4: upload<2GB + gdrive/bucket ? detect+stitch on Cloud Run ? reel via signed URL; ****DEPLOY_REPORT**** (posterity, sample runs + contact sheet) | M6 | ? owner (spend) | ? | ? |

123:| M8 | Per-job ****cost ledger**** (v6 migration: `owner\_id, gpu\_active\_seconds, wall\_seconds, bytes\_out, cost\_usd, cost\_breakdown\_json`) + ****pricebook**** + aggregation + `/api/.../cost` | M7 | ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission ? owner/follow-up) |

124:| M9 | ****Edge auth**** (Cf-Access extraction) + per-tenant scoping on `user\_id`; DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? |

[#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT verify + owner_id tag, default-OFF ?; ****+ exposure-safety boot posture**** in `startup.py` ? edge-auth-on requires the `allowed\_video\_root` jail + CF team/AUD or the server refuses to start, see changelog; consent cols + ToS/DPA ? owner/counsel) |

126:| ****M11**** | ****Data-flywheel harness (D12):**** on a consented adult job, ****capture**** the upload + Gemini reel/rally output ? ****reliably store**** under the per-video workspace ? ****shadow-eval**** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? ****promote**** good ones to the golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the owned model climbs to the D11 floor. Reuses `data\_pipeline/` + `backend/eval/` + the consent gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | ? |

127:| ****M12**** | ****gdrive-api**** (OAuth) StorageBackend (D14 Northstar): ****cloud reads the source from + writes the stitched reel back to the user's Google Drive next to the source**** (per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? | ? |

149:- [] ****truly-local**** (M5) and ****cloud-serving**** (M7) each have a committed ****runlog / DEPLOY_REPORT**** with a real sample run.

155:****ADR lineage:**** ADR-005/008/009/010/011/012/013 ? ****ADR-014**** (`docs/archives/decisions/DECISIONS.md`). ****Strategy:**** `PLATFORM\_ARCHITECTURE.md`, `HOSTING\_PLAN.md`, `VIDEO\_LOCALITY\_MODEL.md`, `DESKTOP\_APP\_PLAN.md`, `LOCAL\_APPLIANCE\_ARCHITECTURE.md` (khelsutra). ****Archived predecessor:**** `docs/archives/past\_projects/DECOUPLED\_COMPUTE/` (esp. `05-COST-ATTRIBUTION.md`, `07-COMPUTE-ABSTRACTION.md`, `06-RISKS-AND-OPEN-QUESTIONS.md`, `DEPLOY\_REPORT\_GCP\_2026-06-20.md`). ****Code anchors [verified]:**** `backend/pipeline/detectors/base.py:164`, `tracknet\_runner.py:281`, `stitcher/compiler.py:303`, `segmenters/trajectory\_hybrid.py:65,237`, `backend/storage/`, `backend/worker/\_\_main\_\_.py:88`, `backend/api/{uploads,job\_worker}.py`, `backend/infrastructure/database.py`. ****Design provenance:**** workflow `wf\_56274ff0` (4 code-mappers ? synthesis; 58 findings).

165:- 2026-06-21 ? ****REAL M7 cloud GPU run ?**** (owner-authorized, within the D17 cap): provisioned an ****L4**** (`g2-standard-4`, asia-southeast1-a; Mumbai stocked-out) ? native WASB on the L4 ? ****22 rallies**** on the smoke clip ?

`gs://?/outputs/05e0e577?/{intervals.csv,report.json}`; VM DELETED (\$0, ~\$0.21). ? the no-Gemini cloud run needs `--set indexing.skip\_ai\_handoff=true` (the fully-local path emits WASB candidate windows directly ? the long-flagged "0 rallies" cold-start fix). The reel ****stitch**** validated locally (`VideoCompiler` ? 22-segment highlights, 85 MB). DEPLOY_REPORT runlog = the cloud-GPU dry-run guide (`docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`).

168:- 2026-06-21 ? ****M6 code landed**** ([#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)): `backend/compute/` ComputeBackend registry ? `CloudRunCompute` dispatches a Cloud Run ****Job**** (injectable client) then ****bucket-polls**** a done-marker via `execution\_policy.poll\_until` + the storage layer (D16); `get\_compute\_backend` returns a backend

```
ONLY for `compute_target=cloud_run` (else `None` ? the worker's in-process path, **default-  
OFF**). The dispatched container runs **inline** (forced `local_gpu`+native, never re-dispatch);  
the worker writes a `--done-uri` completion marker on success.  
`deploy/cloudrun/{Dockerfile,README}` = the L4 image (CUDA+ffmpeg+fonts+`wasb` conda env;  
weights staged per WASB-C3) + the M7 runbook. Adversarial review (`wf_87b22fa4`) folded a  
**blocker** (cloud `fetch` needs a dst ? reads silently returned `{}` on gcs/s3) + a poll-wait  
test gap + `PolicyTimeout`?clean rc 4. Suite green, ruff+mypy clean. **? M6 ? ? the real  
build/push/L4 run + DEPLOY_REPORT is M7 (owner, D17 budget).**  
=== CLOUD_GPU_DRYRUN guide present (SSH-only path)? ===  
100644 blob 8e263b489ecbab4ab7fc73a281459b7b7e203d7f docs/CLOUD_GPU_DRYRUN_GUIDE.md
```

23. assistant (2026-06-22T01:57:00.584Z)

Now I have a very clear picture. Let me verify two more things: the actual SPA source for the
upload?poll?reel flow + whether a compute picker exists, and confirm the M7 status nuance (the
"real L4 run" was a manual SSH run, not the M7 Cloud Run *service* deploy).

24. user (2026-06-22T01:57:03.014Z)

```
=== web/src tree ===  
web/src/api/client.test.ts  
web/src/api/client.ts  
web/src/api/locale.test.ts  
web/src/api/types.ts  
web/src/api/upload.test.ts  
web/src/App.tsx  
web/src/components/a4.touch.test.tsx  
web/src/components/a8.status.test.tsx  
web/src/components/build-error.test.tsx  
web/src/components/CompileResult.tsx  
web/src/components/IngestPanel.test.tsx  
web/src/components/IngestPanel.tsx  
web/src/components/LanguageSwitcher.tsx  
web/src/components/QueryBar.examples.test.ts  
web/src/components/QueryBar.test.tsx  
web/src/components/QueryBar.tsx  
web/src/components/RallyGrid.tsx  
web/src/components/SelectionFooter.test.tsx  
web/src/components/SelectionFooter.tsx  
web/src/components/StatusChip.tsx  
web/src/components/TapFilters.test.tsx  
web/src/components/TapFilters.tsx  
web/src/components/upload.test.tsx  
web/src/hooks/useIngestJob.test.ts  
web/src/hooks/useIngestJob.ts  
web/src/hooks/useRallySelection.ts  
web/src/hooks/useUpload.test.ts  
web/src/hooks/useUpload.ts  
web/src/i18n/catalog-invariants.test.ts  
web/src/i18n/hygiene.test.ts  
web/src/i18n/i18n.test.tsx  
web/src/i18n/index.ts  
web/src/index.css  
web/src/lib/errors.test.ts  
web/src/lib/errors.ts  
web/src/lib/errorText.test.ts  
web/src/lib/errorText.ts  
web/src/lib/floor.test.ts  
web/src/lib/floor.ts  
web/src/lib/format.test.ts  
=== poll/process/compute in src ===  
web/src/api/client.ts:82: * processing. POST /api/process {video_path} ? 202 {job_id, ...}  
[main.py:260]. Fast-fails 400  
web/src/api/client.ts:94:/** Poll a submitted job. GET /api/jobs/{job_id} [main.py:303]. 404 ?  
unknown job_id. */
```

```

web/src/api/client.ts:143:// bodies ~100 MB), then `complete` returns a server-host `path` to
feed straight into /api/process.
web/src/api/locale.test.ts:2:// the reel watermark) and /api/process (PMF analytics). Optional +
omitted when absent, so existing
web/src/api/locale.test.ts:37:   const f = ok({ job_id: "j", status: "queued", poll:
"/api/jobs/j" });
web/src/api/types.ts:48:/** POST /api/process ? 202 [backend/main.py:260]. The job is queued;
poll GET /api/jobs/{job_id}. */
web/src/api/types.ts:53:  /** Relative poll path, e.g. "/api/jobs/<id>". */
web/src/api/types.ts:58:  * ? max_part_mb, then `complete` returns a server-host `path` to feed
straight into POST /api/process. */
web/src/api/types.ts:71:  /** Absolute path on the engine host ? pass as video_path to POST
/api/process. */
web/src/api/types.ts:77:/** GET /api/jobs/{job_id} [backend/main.py:303]. Two status layers:
web/src/components/a8.status.test.tsx:72:   if (u === "/api/process" && (init?.method ??
"GET") === "POST") return process;
web/src/components/a8.status.test.tsx:73:   if (u.startsWith("/api/jobs/")) {
web/src/components/IngestPanel.test.tsx:22:   if (u === "/api/process" && (init?.method ??
"GET") === "POST") return process;
web/src/components/IngestPanel.test.tsx:23:   if (u.startsWith("/api/jobs/")) {
web/src/components/IngestPanel.test.tsx:53:   fetchSeq(httpOk(202, { job_id: "j", status:
"queued", poll: "/api/jobs/j" })), [
web/src/components/IngestPanel.test.tsx:4:// the SAME back-half: POST /api/process ? poll GET
/api/jobs/{id} ? hand the video_id to App. Honest
web/src/components/upload.test.tsx:4:// server PATH is fed to POST /api/process ? poll ?
onLoaded(video_id). The keystone for alpha users
web/src/components/upload.test.tsx:5:// who have a FILE, not a gs:// URI. Also pins that the
file picker is present + labelled (a11y/A4).
web/src/components/upload.test.tsx:33:   if (u === "/api/process" && m === "POST") {
web/src/components/upload.test.tsx:35:   return ok({ job_id: "j", status: "queued", poll:
"/api/jobs/j" });

```

25. assistant (2026-06-22T01:57:10.031Z)

This confirms it definitively. The SPA ingest/progress flow (upload ? `/api/process` ? poll `/api/jobs/{id}` ? reel) is **fully built** (`IngestPanel.tsx`, `useIngestJob.ts`, `useUpload.ts`, with tests). The NEXT_STEPS P0 item #0(ii) "SPA ingest/progress screen (khelsutra web B-P2)" is **stale** ? it shipped (#56 + T4 #66).

There is **no compute picker** in the SPA (no "your machine vs our cloud" / `compute\_target` UI ? that part of the UI-driven gap (D13/D15) is genuinely unbuilt. And the alpha is deployed (per task context) using the CPU droplet + Gemini path, not the M7 Cloud-Run-GPU service.

I have everything I need. Let me compose the structured output.

26. user (2026-06-22T01:57:55.579Z)

Structured output provided successfully

27. assistant (2026-06-22T01:58:00.585Z)

Done. Verified the engine serving/platform/owned-model track against origin/master and the CDS \$7 tracker, returned prioritized next items with owner-gating flags, and corrected five stale doc claims (the biggest being that DEFERRED_HARDENING #16 async-jobs is already DONE, not the next slice, and that the SPA ingest screen already shipped ? the real UI gap is only the compute picker).